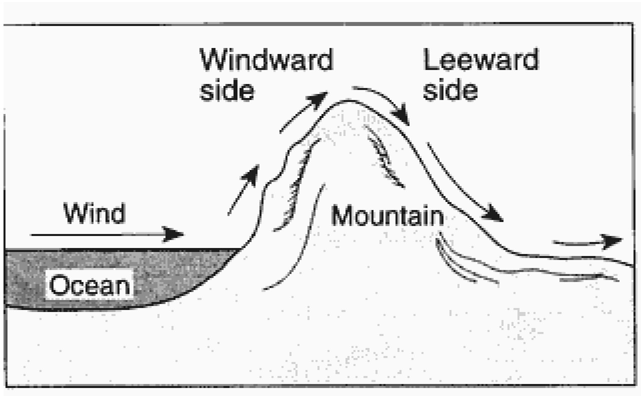
Cap-One-Haiti
Rain Shadow

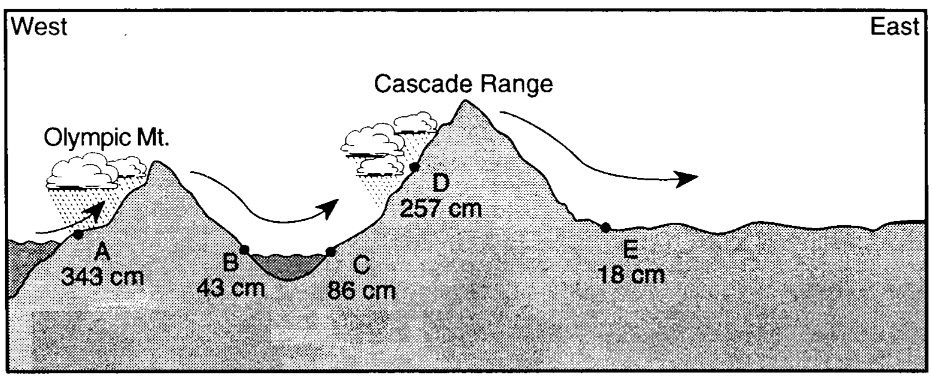
1. The diagram below shows the flow of planetary winds over a mountain ridge.



As air rises on the windward side of the mountain ridge, the air's temperature decreases. Which process usually causes this temperature decrease?

A) expansion of rising air
B) compression of rising air
C) precipitation from clouds
D) evaporation from clouds

2. The diagram below shows the average yearly precipitation, in centimeters, at locations *A* through *E* across the State of Washington. Arrows indicate the direction of prevailing winds.

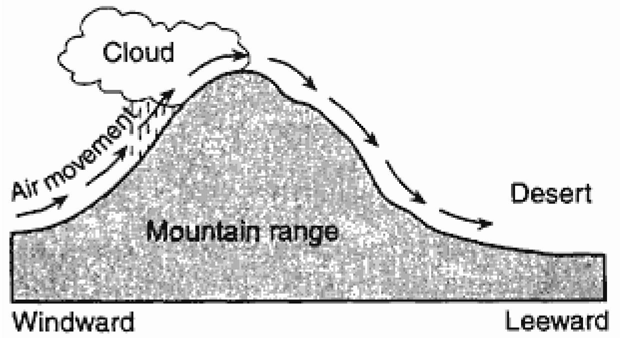


(Not drawn to scale)

Which statement best explains why location *B* and location *E* receive relatively low average yearly precipitation?

A) These locations are on the leeward side of mountain ranges.
B) These locations are on the windward side of mountain ranges.
C) These locations receive more insolation than the other locations.
D) These locations receive less insolation than the other locations.

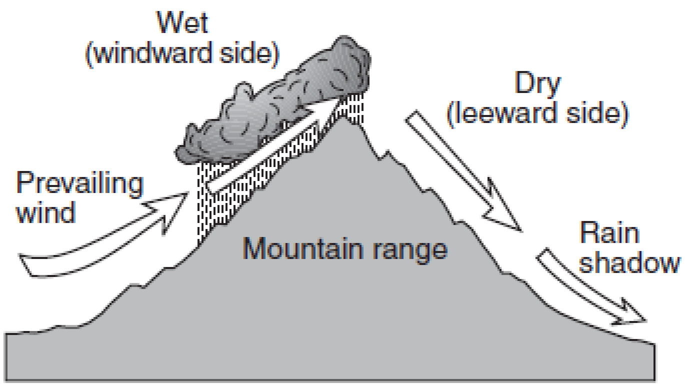
3 A desert often forms on the leeward side of a mountain range, as shown in the cross section below.



After most of the moisture is removed from the air on the windward side, deserts form on the leeward side because the sinking air

A) compresses and warms
B) compresses and cools
C) expands and warms
D) expands and cools

4 The cross section below shows how prevailing winds have caused different climates on the windward and leeward sides of a mountain range.



Why does the windward side of this mountain have a wet climate?

A) Rising air compresses and cools, causing the water droplets to evaporate.
B) Rising air compresses and warms, causing the water vapor to condense.
C) Rising air expands and cools, causing the water vapor to condense.
D) Rising air expands and warms, causing the water droplets to evaporate.

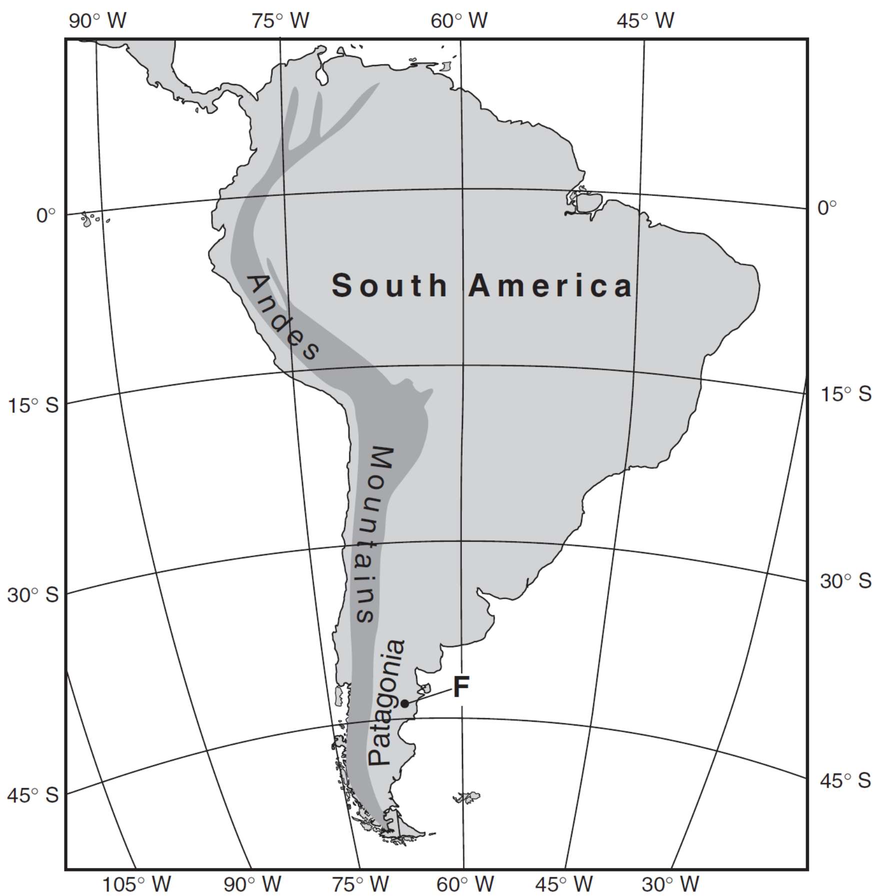
5 Base your answer to the following question on the passage and map below. Point *F* on the map shows the location where an unusual mammal fossil was found.

Fossil Jaw of Mammal Found in South America

Paleontologists working in Patagonia have found the tiny fossil jaw that may be the first evidence of early mammals in South America.

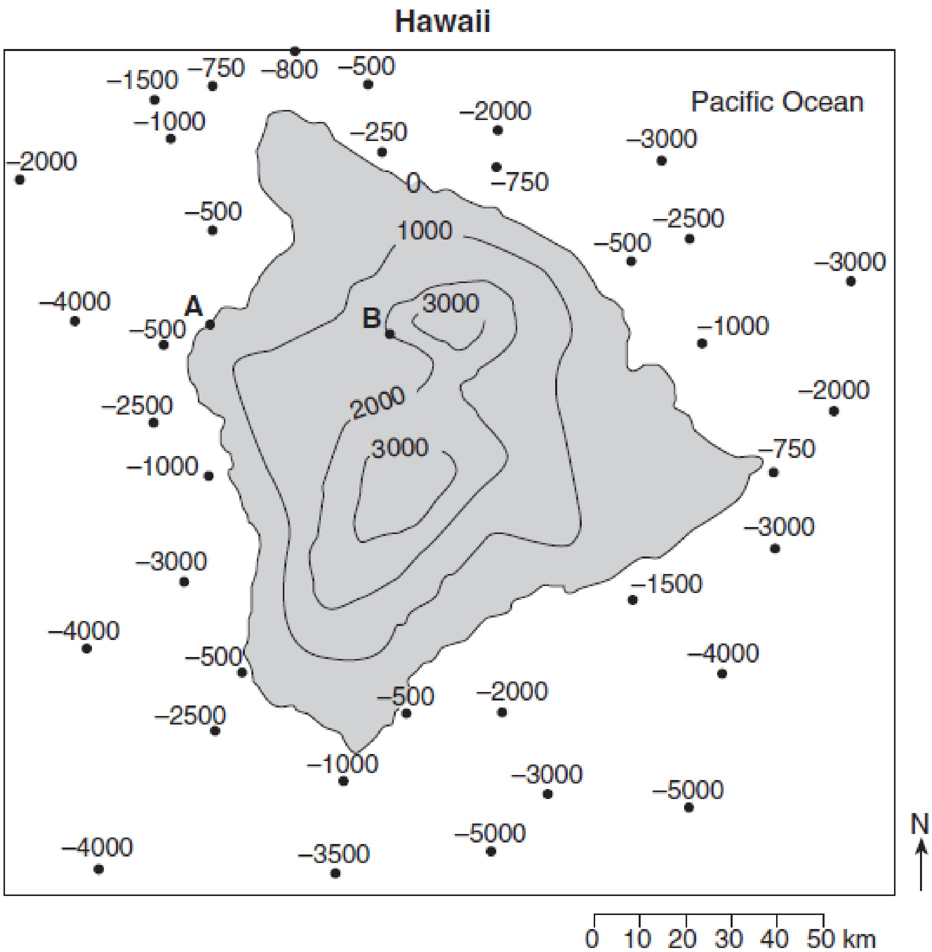
The fossil, which measures less than a quarter-inch long, is believed to be from the middle or late Jurassic Period. Researchers said it suggests that mammals developed independently in the Southern Hemisphere.

The fossil, named *Asfaltomylos patagonicus*, was discovered in a shale formation in Patagonia. Dinosaurs were the dominant land animal at that time. Mammals were tiny, and hunted insects in the dense tropical vegetation. The now-arid region also has yielded some remarkable dinosaur fossils from the same period in a vast ancient boneyard covering hundreds of square miles.

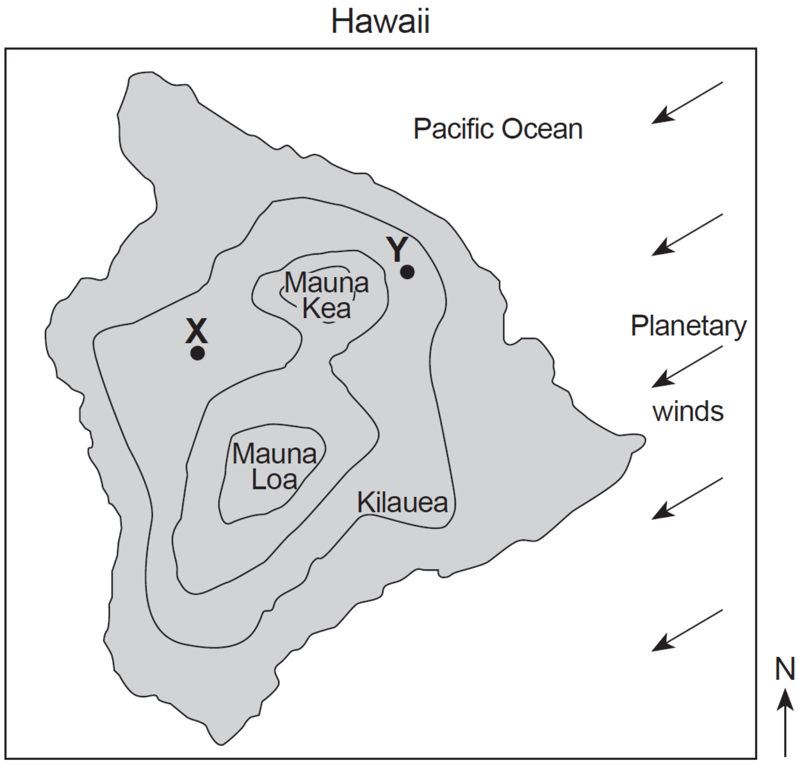


Explain how the uplift of the Andes Mountains changed eastern Patagonia's climate from a wet tropical forest at the time *Asfaltomylos patagonicus* lived to the arid conditions of today.

6 Base your answer to the following question on the topographic map of Hawaii below and on your knowledge of Earth Science. Points *A* and *B* represent surface locations on the island. Land elevations and Pacific Ocean depths are shown in meters.



The map below shows the locations of three volcanoes on the island of Hawaii. The arrows represent the direction of the planetary winds. Points *X* and *Y* represent surface locations on the island.



Explain why location *X* usually receives *less* annual precipitation than location *Y*.